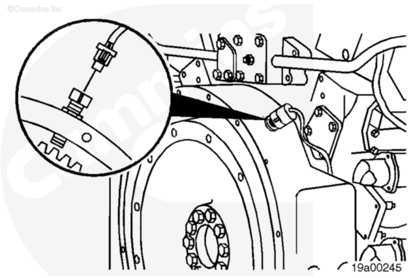
Trainings

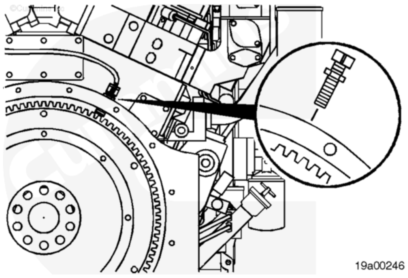
Remove

Disconnect the engine speed sensor connectors from the engine harness.



Loosen the capscrew.

Remove the engine speed sensor from the flywheel housing.

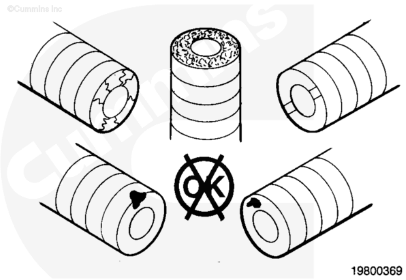


Inspect for Reuse

Inspect the engine speed sensor for debris, cracked or chipped potting, extruded potting, and damage from contact with the flywheel.

If there is debris on the engine speed sensor, clean the sensor.

If the sensor is chipped, cracked, extruded, or damaged, replace the sensor with a new one.



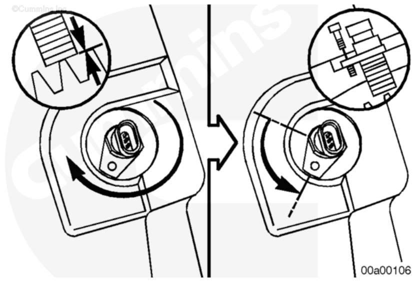
Install

Note : Check that the engine speed sensor is directly above a flywheel ring gear tooth.

Install the engine speed sensor until it contacts the flywheel. Back the sensor off until the lock hole is aligned.

Install the lock capscrew and tighten.

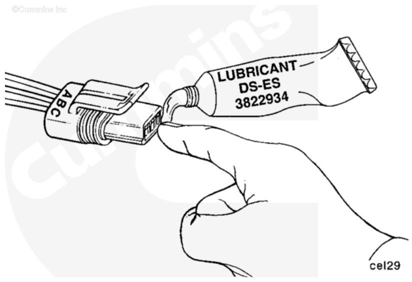
Torque Value: 7 n•m [62 in-lb]



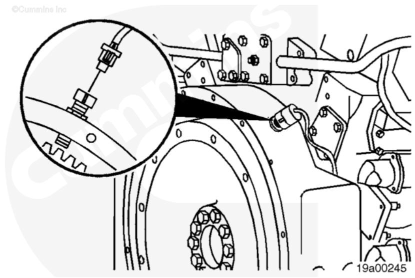
⚠ CAUTION ⚠

Use only Cummins-recommended lubricant DS-ES, Part Number 3822934. Other lubricants, such as lubricating oil or grease, in the connectors can cause ECM damage, poor engine performance, or premature connector pin wear.

Apply a small amount of lubricant to the connector terminals. Before installing, fill the entire connector cavity with lubricant.



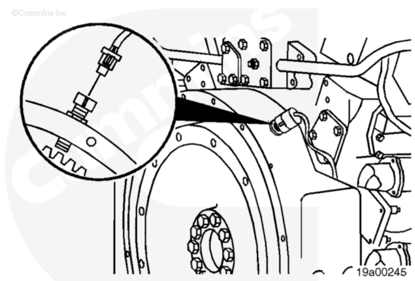
Connect the sensor to the sensor harness. Push the connectors together until they lock.



Resistance Check

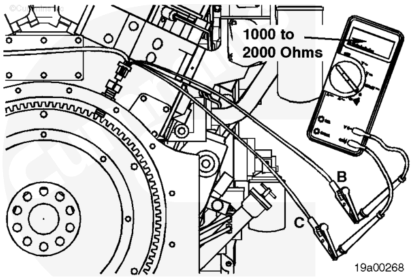
Separate the three-way connector. Lift the tab and pull the connector apart. Install a mating connector with short test leads on the sensor connector.

Note : The purpose of installing a mating connector is to allow the electrical leads of the sensor to be softly flexed to check for damaged or partially broken wire strands under the insulation.



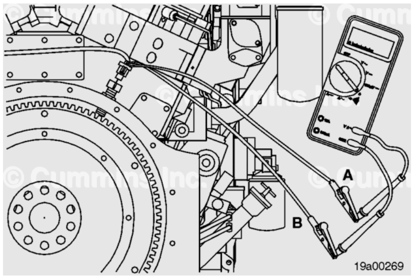
Use a multimeter to measure the resistance from pin C to pin B.

The multimeter **must** show between 1000 and 2000 ohms.



Measure the resistance from pin A to pin B. The multimeter **must** measure between 1000 and 2000 ohms. If both resistance values are within the specifications, the sensor **must** still be checked for short circuit to ground.

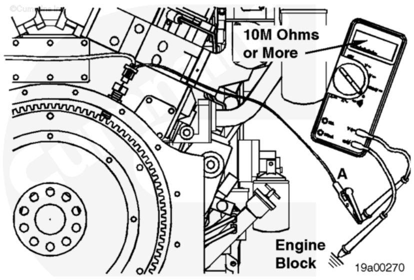
If any of the preceding checks fail, replace the engine speed sensor.



Check for Short Circuit to Ground

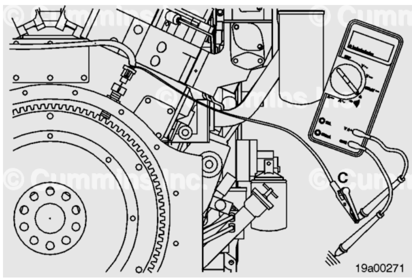
Measure the resistance from pin A to the engine block. The multimeter **must** show an open circuit (10M ohms or more).

Note : The open circuit specification (10M ohms) for the engine speed sensor is higher than the open circuit specification used throughout the manual due to the sensitivity of the engine speed sensor.



Measure the resistance from pin C to the engine block. The multimeter **must** show an open circuit (100k ohms or more).

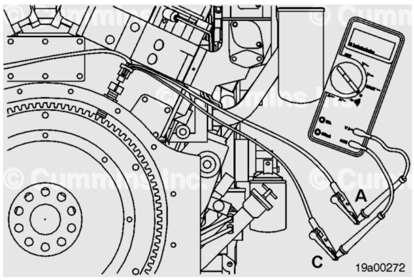
If either of the preceding tests fails, replace the engine speed sensor.



Check for Short Circuit from Pin to Pin

Measure the resistance from pin A to pin C. The multimeter **must** show an open circuit (100k ohms or more). If any of the previous resistance checks are **not** within specifications, the sensor has failed.

Replace the sensor.



Last Modified: 10-Feb-2003